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6a. Write a Python program to import a CSV dataset using Pandas and display the first 5 and last 5 records

```
import pandas as pd
```

```
df = pd.read_csv("Students.csv")
```

```
print(df.head())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Sujal	MCOM	78	85
1	102	Sakashi	MA	65	90
2	103	Jueli	BSc IT	88	92
3	104	Neha	LLB	72	80
4	105	Rohan	BSc IT	55	70

6b. Write a Python program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information.

```
print(df.shape)
```

```
(5, 5)
```

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

```
print(df.dtypes)
```

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

6c. Write a Python program to filter records from the dataset using conditions.

```
result = df[df["Attendance"]>=90]
print(result)
```

	Roll No	Name	Department	Marks	Attendance
1	102	Sakashi	MA	65	90
2	103	Jueli	BSc IT	88	92

6d. Write a Python program to sort the dataset based on a selected column and display the top 5 records.

```
result = df.sort_values("Marks")  
print(result)
```

	Roll No	Name	Department	Marks	Attendance
4	105	Rohan	BSc IT	55	70
1	102	Sakashi	MA	65	90
3	104	Neha	LLB	72	80
0	101	Sujal	MCOM	78	85
2	103	Jueli	BSc IT	88	92

6e. Write a Python program to perform data analysis on the dataset by finding the maximum, minimum, average, and total values of a numerical column.

```
print("Maximum Marks ",df["Marks"].max())  
print("Minimum Marks ",df["Marks"].min())  
print("Average Marks",df["Marks"].mean())  
print("Total Marks",df["Marks"].sum())
```

```
Maximum Marks  88  
Minimum Marks  55  
Average Marks  71.6  
Total Marks  358
```